

HASTELLOY B-2 (UNS N10665)

Description:

Hastelloy B-2 is a nickel-molybdenum alloy designed for exceptional resistance to reducing acids (e.g., hydrochloric, sulfuric, acetic, phosphoric). Its low chromium, carbon, and silicon content ensures minimal carbide formation during welding, excellent ductility, and reliable corrosion resistance under moderate to high temperatures. Widely used in chemical processing, pickling, pharmaceutical, and environmental applications.

Chemical Composition

Element	Composition (%)
Nickel	65 (Balance)
Chromium	1.0
Molybdenum	26–30
Iron	2.0
Cobalt	1.0
Manganese	1.0
Silicon	0.10
Carbon	0.02
Phosphorus	0.04
Sulfur	0.03

Physical Properties

Property	Value
Density	9.24 g/cm ³
Melting Point	1370°C (2500°F)
Tensile Strength	110,000 psi / 760 MPa
Yield Strength	52,000 psi / 355 MPa
Elongation	40%

Mechanical Properties

Property	Metric	Imperial
Tensile Strength (@538°C)	760 MPa	110,000 psi
Yield Strength (@427°C)	355 MPa	52,000 psi
Elastic Modulus (RT)	210 GPa	30,000 ksi
Elongation at Break (@204°C)	40%	40%
Hardness, Rockwell B (plate)	88	88

Available Products

- Pipes: Seamless, ERW & EFW
- Tubes: Capillary, U-Bend
- Sheets & Plates
- Coils & Strips
- Bars: Round, Flat, Hex, Square
- Pipe Fittings: Elbows, Tees, Reducers, Caps, Stub Ends
- Flanges: Weld Neck, Slip-On, Blind, Socket Weld, Threaded
- Fasteners: Bolts, Nuts, Screws, Washers, Studs
- Custom Fabricated Components

Key Advantages

- Exceptional resistance to reducing acids
- Maintains integrity and strength at elevated temperatures
- Minimizes carbide formation during welding
- Excellent ductility and weldability
- Reliable for acid handling equipment, tanks, piping, and heat exchangers

Applications

- **Chemical Processing:** Reactors, heat exchangers, distillation columns, piping systems handling reducing acids
- **Pickling & Acid Regeneration:** Pickling lines, acid tanks, regeneration units
- **Agrochemical & Fertilizer:** Acid handling systems, reactors, piping
- **Pharmaceutical & Fine Chemicals:** Vessels and systems needing high purity and corrosion resistance
- **Electronics & Semiconductor:** Acid handling components in wet etching and cleaning
- **Waste Treatment & Pollution Control:** Acidic waste stream and off-gas treatment systems